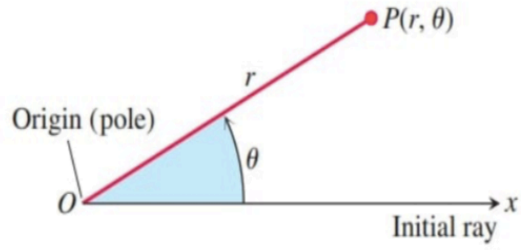
10.3 Polar Coordinates

Cartesian coordinate system (The ***rectangular coordinate system***)

- A Cartesian coordinate system specifies a point in the plane by an ordered pair of numbers called ***Cartesian coordinates*** (or ***coordinates***). These coordinates give the directed distances from two perpendicular axes.
- The same principle is used to specify the position of any point in **three-dimensional space** by an ordered triplet of numbers which are signed distances to three mutually perpendicular planes.
- Any point can be located by a **unique** ordered pair (triplet) of numbers.

Polar coordinate system

- Polar coordinates offer an alternative way of locating points in a plane. These coordinates are useful because for certain types of regions and curves, polar coordinates provide very simple descriptions and equations.
- A polar coordinate system specifies a point in the plane by means of coordinates **relative to a fixed point *O* (called the pole) and an initial axis from *O* (called the initial ray).**

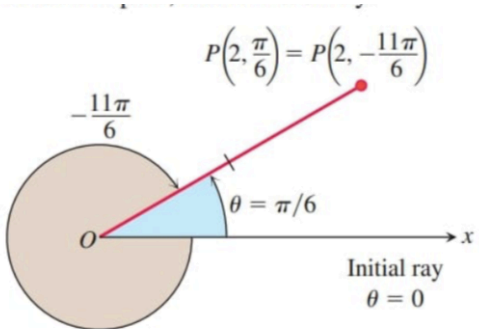


Polar Coordinates ***P*(*r*, θ)**

r gives the **directed distance from *O* to *P***,
θ gives the **directed angle from the initial ray to *OP***.
(θ is positive when measured counterclockwise and is negative when measured clockwise.)

- Polar coordinates of a point are ***NOT unique***.

Example: $\left(2, \frac{13\pi}{6}\right), \left(2, \frac{25\pi}{4}\right), \left(2, -\frac{11\pi}{6}\right)$ are some polar coordinates for the point $P\left(2, \frac{\pi}{6}\right)$.



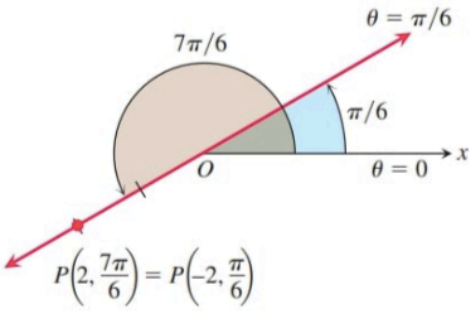
- When $r = 0$, **(0, θ)** are polar coordinates for **the pole** for all values θ.

  Example: (0, 0), (0, π), (0, π/4), (0, −π/2), (0, −5π) are some polar coordinates for the pole.

- When ***r* is negative**, the point P can be reached by turning θ radian from the initial ray and going backward $|r|$ units.

> ❖ $P(r, \theta) = P(-r, \theta + \pi) = P(-r, \theta - \pi)$

Example: $\left(-2, \frac{\pi}{6}\right)$ and $\left(-2, -\frac{5\pi}{6}\right)$ are also polar coordinates for the point $P\left(2, \frac{7\pi}{6}\right)$.



- A point in the plane (space) has just 1 pair (triplet) of Cartesian coordinates, but it has infinitely many pairs of polar coordinates.

> ❖ **$(r, \theta + 2n\pi)$ and $(-r, \theta - \pi + 2n\pi)$, $n \in Z$, are polar coordinates of the same point.**

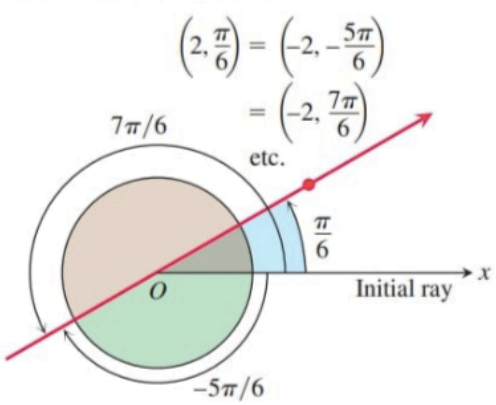
Example 1: Find all polar coordinates of the point $\left(2, \frac{\pi}{6}\right)$.

Solution:

For $r = 2$, the angles are $\frac{\pi}{6}, \frac{\pi}{6} \pm 2\pi, \frac{\pi}{6} \pm 4\pi, \frac{\pi}{6} \pm 6\pi, \cdots$.

For $r = -2$, the angles are $-\frac{5\pi}{6}, -\frac{5\pi}{6} \pm 2\pi, -\frac{5\pi}{6} \pm 4\pi, -\frac{5\pi}{6} \pm 6\pi, \cdots$.

All polar coordinates of the point $\left(2, \frac{\pi}{6}\right)$ are $\left(2, \frac{\pi}{6} + 2n\pi\right)$ and $\left(2, -\frac{5\pi}{6} + 2n\pi\right)$ where n is an integer.



Example 2: Plot the points with polar coordinates $P\left(2, \frac{4\pi}{3}\right)$, $Q\left(2, -\frac{\pi}{4}\right)$, $R\left(-1, \frac{\pi}{2}\right)$, $S(-1, -\pi)$ and $T\left(-2, -\frac{\pi}{6}\right)$. Find all polar coordinates of the point S and the point T.

Polar Equations and graphs

- The polar equation $r = a$ represents **the circle of radius $|a|$ centered at O.**
- The polar equation $\theta = \theta_0$ represents **the line through O making an angle θ_0 with the initial ray.**

Example 3:

(a) $r = 4$ and $r = -4$ are polar equations for the circle of radius 4 centered at the pole O.

(b) $\theta = \pi/3$, $\theta = 4\pi/3$ and $\theta = -2\pi/3$ are polar equations for the line through O making an angle $\pi/3$ with the initial ray.

Example 4:

Graph the set of points whose polar coordinates satisfy the following conditions.

(a) $1 \le r \le 3$ and $\pi/2 \le \theta \le \pi$.

(b) $-1 \le r \le 2$ and $\theta = \pi/3$.

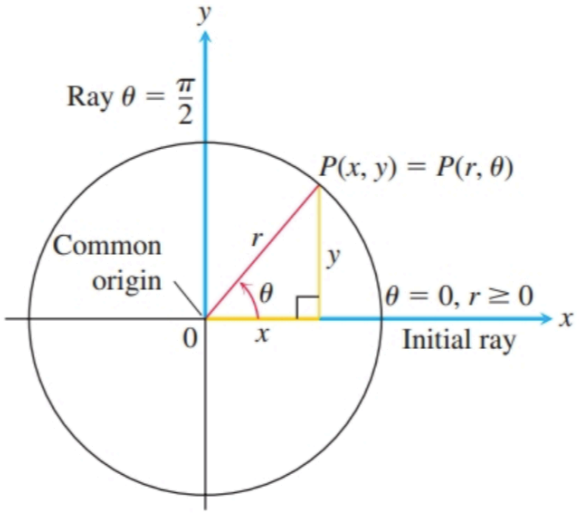
(c) $r \le -1$ and $0 \le \theta \le \pi/4$.

(d) $2 \le r \le 3$.

(e) $-\pi/2 \le \theta \le \pi/3$.

Equations Relating Polar and Cartesian Coordinates

$$x = r\cos\theta \ , \quad y = r\sin\theta \ , \quad x^2 + y^2 = r^2 \quad \text{and} \quad \tan\theta = \frac{y}{x}, \ x \neq 0.$$



Example 5:
Here are some plane curves expressed in terms of both polar coordinate and Cartesian coordinate equations.

Polar equation	Cartesian equivalent
$r\cos\theta = 2$	$x = 2$
$r^2\cos\theta\sin\theta = 4$	$xy = 4$
$r^2\cos^2\theta - r^2\sin^2\theta = 1$	$x^2 - y^2 = 1$
$r = 1 + 2r\cos\theta$	$y^2 - 3x^2 - 4x - 1 = 0$
$r = 1 - \cos\theta$	$x^4 + y^4 + 2x^2y^2 + 2x^3 + 2xy^2 - y^2 = 0$

Some curves are more simply expressed with polar coordinates; others are not.

Example 6: Find a polar equation for the circle $x^2 + (y - 3)^2 = 9$.

Solution:

$$
\begin{aligned}
& x^2 + (y-2)^2 = 9 \\
\Rightarrow \quad & x^2 + y^2 - 6y + 9 = 9 \\
\Rightarrow \quad & x^2 + y^2 - 6y = 0 \\
\Rightarrow \quad & r^2 - 6r\sin\theta = 0 \\
\Rightarrow \quad & r(r - 6\sin\theta) = 0 \\
\Rightarrow \quad & r = 6\sin\theta \qquad \text{since } r \neq 0.
\end{aligned}
$$

Example 7: Replace each of the following Cartesian equations by equivalent polar equation.

(a) $(x-1)^2 + (y+2)^2 = 9$

(b) $x^2 - 3xy + y^2 = 1$

Example 8: Replace each of the following polar equations by equivalent Cartesian equation.

(a) $r\cos\theta = 4$

(b) $r^2 = 4r\cos\theta$

(c) $r = \dfrac{4}{2\cos\theta - \sin\theta}$

(d) $\cos^2\theta - \sin^2\theta = 1$

(e) $r = 1 + 2r\cos\theta$

(f) $r = 1 - \cos\theta$